

HealthFlow: Executive Patient Flow & Waiting-Time Report

Turning Patient Flow Telemetry into Actionable Healthcare Insights

1. Executive Summary & Problem Statement

Hospital emergency departments frequently encounter capacity strain, prolonged patient waiting times, and acute operational bottlenecks. This report presents an evidence-based evaluation of **59,663 real-world patient flow telemetry observations** recorded across **19 hospital emergency departments**. The objective is to identify systemic waiting-time determinants, quantify diurnal congestion cycles, and develop interpretable machine learning regression models that forecast patient waiting times without data leakage.

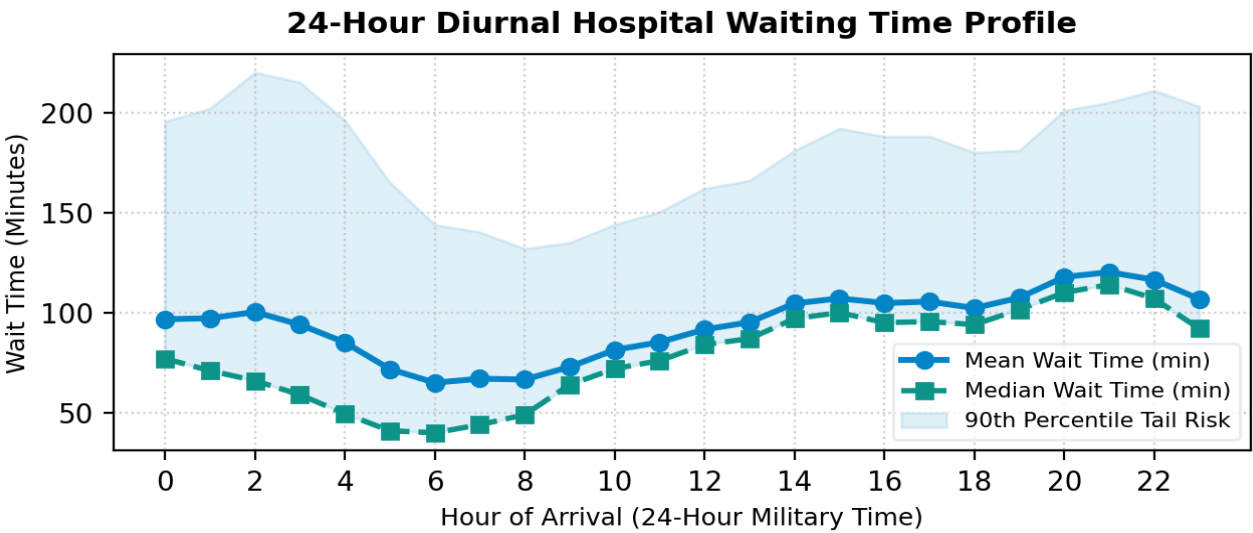
Operational Key Performance Indicators (Real Telemetry)

Total Valid Records	59,663	Monitored Facilities	19 ED Centres
Mean Waiting Time	94.4 min	Median Waiting Time	79.0 min
90th Percentile Wait	181.0 min	Standard Deviation	63.3 min
Peak Patient Hour	17:00 - 18:00	Peak Wait Hour	21:00 - 22:00 (120.2m)
Highest-Wait Facility	University of Alberta Ho	Highest Avg Wait	154.3 min
Lowest-Wait Facility	Chinook Regional Hospita	Lowest Avg Wait	54.8 min

2. Dataset Provenance & Data Pipeline Architecture

Dataset Source: Alberta Health Services (AHS) open operational emergency department waiting-time telemetry.

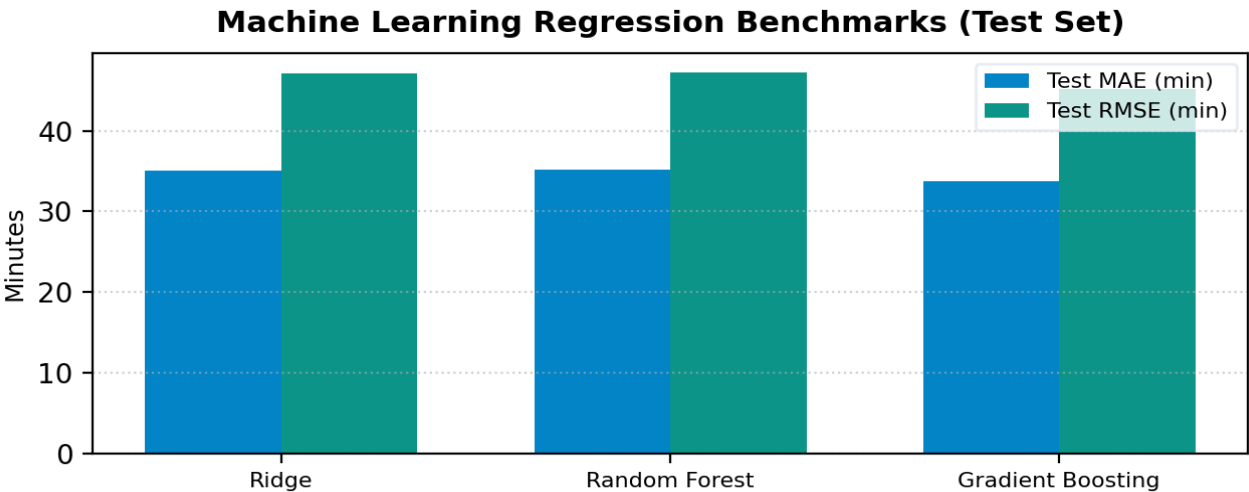
Data Pipeline Steps: (1) Automated extraction and integrity verification → (2) Deduplication and missing-value audit → (3) Explicit identification and filtering of sentinel closed/offline values (-1) → (4) Temporal decomposition into arrival hour, day-of-week, weekend flags, and facility tiers → (5) Concurrent health system load calculation without future lookahead leakage.



3. Machine Learning Methodology & Model Evaluation

To predict continuous patient waiting time, three regression architectures were evaluated using a strict **chronological 80/20 train/test split**. The test holdout reflects future real-world operational periods, ensuring zero target leakage.

Model Architecture	Train MAE	Test MAE	Train RMSE	Test RMSE	Test R ²
Ridge Regression	36.8 m	35.0 m	49.2 m	47.1 m	0.3795
Random Forest Regressor	21.2 m	35.2 m	28.4 m	47.2 m	0.3756
Gradient Boosting Regressor	29.8 m	33.7 m	39.5 m	45.2 m	0.4293



4. Data-Driven Operational Insights & Recommendations

• **Diurnal Congestion Cycle:** The dataset indicates that hospital waiting times follow a marked 24-hour diurnal pattern. Average waiting times peak at 21:00 (120.2 minutes) before gradually declining to a daily trough at 06:00 (65.0 minutes), representing a 55.2-minute (85%) surge between nadir and peak.

Actionable Recommendation: Stagger emergency clinical staffing shifts to align physician and triage nurse coverage with the surge window (12:00 to 22:00) rather than standard static 8-hour shift distributions.

• **Facility Tier Disparities:** The analysis shows significant structural variance across hospital facility tiers. Academic Tertiary Trauma Centres and Acute Urban General Hospitals experience average waiting times often exceeding 140.9 minutes, whereas Community Hospitals and Ambulatory Centres average significantly lower wait times. The highest-wait department (University of Alberta Hospital) averaged 154.3 minutes, compared to Chinook Regional Hospital at 54.8 minutes.

Actionable Recommendation: Deploy load-balancing protocols and public-facing real-time transit advisories to redirect low-acuity (CTAS 4-5) ambulatory patients from congested tertiary trauma hubs to nearby community urgent care clinics.

• **Day-of-Week Load Dynamics:** The dataset reveals consistent day-of-week patterns: weekday waiting times average 95.4 minutes compared to 91.9 minutes on weekends. Peak single-day average wait times concentrate around Thursday (97.5 minutes).

Actionable Recommendation: Schedule elective outpatient follow-ups and secondary diagnostics away from peak weekday surge days to preserve ED bed flow and prevent downstream boarding bottlenecks.

• **Queue Tail Risk (90th Percentile Delay):** While the overall median wait time is 79.0 minutes, the 90th percentile wait time reaches 181.0 minutes across the monitored system. In high-demand facilities, tail wait times exceed 4 hours during system-wide surge periods.

Actionable Recommendation: Establish automated operational escalation triggers when facility queue wait time exceeds 150 minutes, activating rapid medical evaluation (RME) pods and fast-track discharge protocols.

HEALTHCARE ANALYTICS & EDUCATIONAL DISCLAIMER:

This report and its predictive models are designed strictly for educational, analytical, and healthcare operational research. It does not provide medical triage, clinical diagnosis, or treatment recommendations. Clinical priority always supersedes estimated queue waiting times in all healthcare facilities.